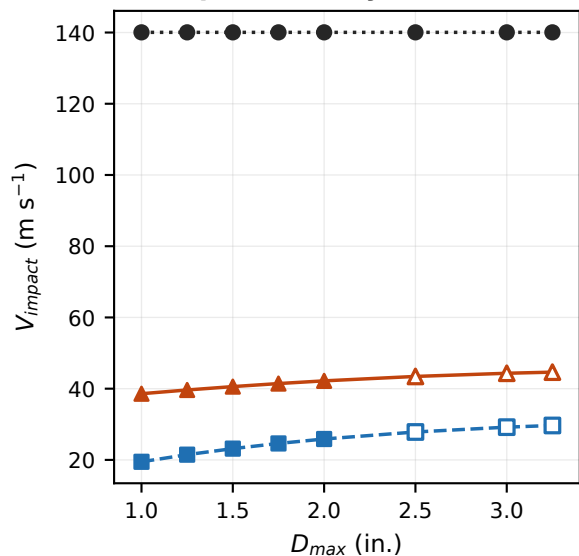
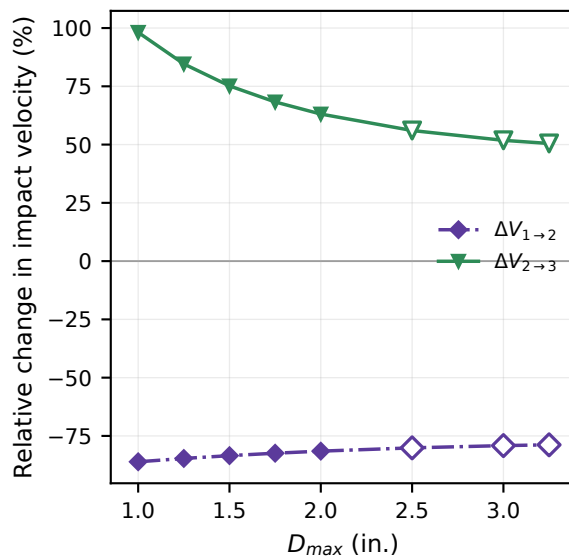


Figure 2. Progressive effect of modeling fidelity on the pre-impact state (Models 1-3)

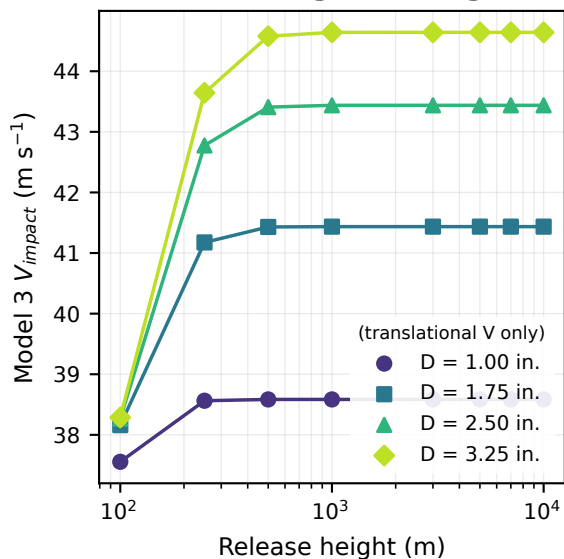
(a) Impact velocity vs. hail diameter



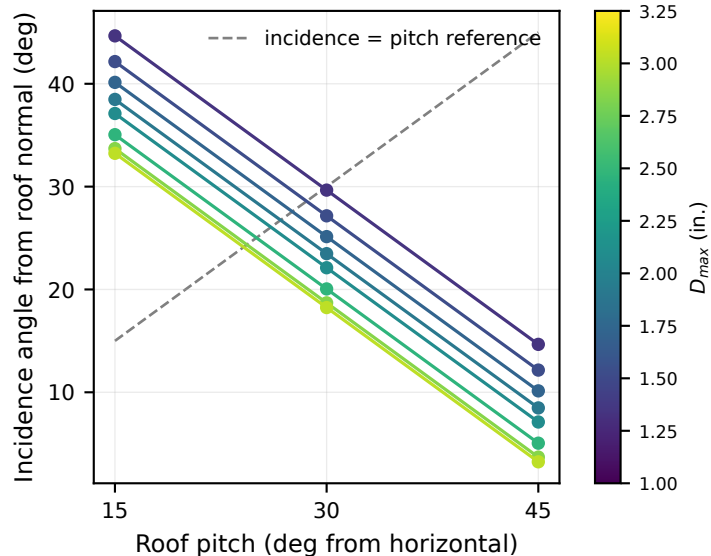
(b) Incremental effect of added fidelity



(c) Release-height convergence — Model 3



(d) Incidence condition — Model 3



●●● Model 1 — baseline (gravity only)
 ■■■ Model 2 — + aerodynamic drag

▲ Model 3 — + environmental fidelity
 ○ Open marker: outside Theis et al. (2026) characterized aerodynamic range